PART IIA, PAPER 3, SUPERVISION 8
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SOLUTIONS TO PROBLEMS

S8.1 Cambridge beating Oxford for the second supervision in a row.
This question is about an alternative way to think about S7.2. Recall that the `STATA` file `BoatRace.dta` contains information on the annual Oxford and Cambridge Boat Race for the period 1946-2013. Variable $\texttt{Winner}_t$ equals 1 if the winner of the race during year t is Cambridge, and 0 if it is the other place. Variable $\texttt{WinTime}_t$ is the winning time (in seconds) in the year t race. Before running any commands, make sure that you declare variable $\texttt{Year}_t$ as time by typing `tsset Year`.

(a) Generate the first differences in $\texttt{WinTime}_t$ and call it $\texttt{DWT}_t$. Use `STATA` to estimate the following AR(1) model for $\texttt{DWT}_t$:

$$\texttt{DWT}_t = \alpha + \beta \texttt{DWT}_{t-1} + \varepsilon_t. \tag{1}$$

Solution: We get the following estimates and standard errors (in parentheses)

$$\texttt{DWT}_t = -\underset{(9.29)}{6.23} - \underset{(0.10)}{0.55}\,\texttt{DWT}_{t-1}, \quad R^2 = 0.32.$$

(b) Suppose that the data generating process for $\texttt{WinTime}_t$ is

$$\texttt{WinTime}_t = \gamma + \delta t + v_t, \tag{2}$$

where v_t is white noise. Argue that in such a case $\texttt{DWT}_t$ can be expressed as

$$\texttt{DWT}_t = \frac{3}{2}\delta - \frac{1}{2}\texttt{DWT}_{t-1} + u_t, \tag{3}$$

where u_t and $\texttt{DWT}_{t-1}$ are not correlated and $\mathbb{E}[u_t] = 0$. What do you think will be the probability limit of your estimate $\hat{\beta}$ from (a)? Explain. [Hint: use equations (2) and (3) to express u_t as a function of v_t and its lags.]
Solution: Based on the regression for $\texttt{WinTime}_t$, its first difference is given by

$$\texttt{DWT}_t = \texttt{WinTime}_t - \texttt{WinTime}_{t-1} = (\gamma + \delta t + v_t) - (\gamma + \delta(t-1) + v_{t-1}) = \delta + v_t - v_{t-1}.$$

Then, from the model (3), the residual is defined as

$$\begin{aligned} u_t &= \texttt{DWT}_t - \frac{3}{2}\delta + \frac{1}{2}\texttt{DWT}_{t-1} \\ &= \delta + v_t - v_{t-1} - \frac{3}{2}\delta + \frac{1}{2}(\delta + v_{t-1} - v_{t-2}) \\ &= v_t - \frac{1}{2}v_{t-1} - \frac{1}{2}v_{t-2} \end{aligned}$$

The error term u_t in the population linear model (3) follows an MA(2) process. Notice that $\mathbb{E}[u_t] = 0$ since $\mathbb{E}[v_t] = 0$ for all t. However, $\text{Cov}(u_t, u_{t-1}) \neq 0$ and so the error in (3) are serially correlated, in particular

$$\text{Cov}(u_t, u_{t-1}) = \text{Cov}\left(v_t - \frac{1}{2}v_{t-1} - \frac{1}{2}v_{t-2}, v_{t-1} - \frac{1}{2}v_{t-2} - \frac{1}{2}v_{t-3}\right)$$

$$
\begin{aligned}
&= -\frac{1}{2}\text{Cov}\left(v_{t-1}, v_{t-1}\right) + \frac{1}{4}\text{Cov}\left(v_{t-2}, v_{t-2}\right) \\
&= -\frac{1}{4}\sigma^2
\end{aligned}
$$

where $\text{Cov}\left(v_t, v_t\right) = \text{Var}\left(v_t\right) = \sigma^2$ for all t. Finally, compute $\text{Cov}\left(u_t, \texttt{DWT}_{t-1}\right)$ to check that the error terms and the explanatory variable in (3) are uncorrelated:

$$
\begin{aligned}
\text{Cov}\left(u_t, \texttt{DWT}_{t-1}\right) &= \text{Cov}\left(v_t - \frac{1}{2}v_{t-1} - \frac{1}{2}v_{t-2}, \delta + v_{t-1} - v_{t-2}\right) \\
&= -\frac{1}{2}\text{Cov}\left(v_{t-1}, v_{t-1}\right) + \frac{1}{2}\text{Cov}\left(v_{t-2}, v_{t-2}\right) \\
&= 0
\end{aligned}
$$

We have just shown that $\texttt{DWT}_{t-1}$ is exogenous with respect to the error term u_i in regression (3). Therefore, the OLS estimator of β in the linear model (1) will be consistent, and since $\beta = -1/2$, conclude that $\hat{\beta} \to -1/2$, as $T \to \infty$.

(c) Your friend suggests that you can test the hypothesis that the true data generating process is the linear model in (3) using the t-test on the hypothesis $H_0\colon \beta = -1/2$ in regression (1). Explain why the default `STATA` version of such a test would not be valid. Can you propose a valid test?

Solution: We have shown that u_t follows an MA(2) process and so is serially correlated. Hence, the typical estimated standard errors of the OLS estimators reported by `STATA` will be invalid. Instead, we could estimate (1) with standard errors robust to serial correlation (HAC) and then conduct the test.

S8.2 Non-stationarity shall not pass! [...the Dickey-Fuler test].

An investigator analysing the annual time series properties of consumers' expenditure and disposable income in the UK over the period 1952-2002 estimates the following three regressions (figures in parentheses are t-statistics).

$$\Delta c_t = \underset{(2.08)}{2.3723} + \underset{(2.0)}{0.0050}t - \underset{(-2.06)}{0.2023}c_{t-1} \tag{4}$$

$$\Delta y_t = \underset{(2.43)}{2.8088} + \underset{(2.32)}{0.0063}t - \underset{(-2.40)}{0.2386}y_{t-1} \tag{5}$$

$$\Delta \texttt{apc}_t = \underset{(-2.34)}{-0.0154} - \underset{(-2.03)}{0.1452}\texttt{apc}_{t-1} \tag{6}$$

where c_t is the log of expenditure at 1990 prices, y_t is the log of personal disposable income at 1990 prices, and $\texttt{apc}_t$ is the log of the average propensity to consume.

(a) Use the Dickey-Fuller test to test the hypothesis that c_t, y_t and $\texttt{apc}_t$ are non-stationary. For each case, justify the choice of deterministic parameters in your tests and state clearly the alternative hypothesis. Can you establish the order of integration of the processes based on the results of your tests? If not, what additional information would you require to do so?

Solution: The general specification of the Dickey-Fuller test with a constant and a trend for consumption and income is:

$$\Delta X_t = \alpha + \delta t + \beta X_{t-1} + \varepsilon_t \tag{7}$$

The number of observations $T = 50$. We are testing $H_0 \colon \beta = 0$ against $H_1 \colon \beta < 0$. The constant and trend specification for consumption and income is chosen since these processes exhibit growing behavior. The Dickey-Fuller critical values at 1% and 5% level are respectively -4.15 and -3.50 for $T = 50$. The test statistic are $t_\beta = -2.06$ for consumption, and $t_\beta = -2.40$ for income. We cannot reject the null of a unit root in either case. Finally, using the model with a constant and no trend for $\Delta\texttt{apc}_t$, the test statistic $t_\beta = -2.03$. For this specification with $T = 50$, the DF critical values at 1% and 5% level are respectively -3.58 and -2.93. We cannot reject the null hypothesis of a unit root. All three time-series are non-stationary. However, we can cannot establish the order of integration based on these tests alone. For example, the process X_t is $I(1)$ if it is non-stationary, but its first difference, ΔX_t, is stationary. Hence, to establish whether the underlying processes are $I(1)$, we would also need to conduct unit root tests for their first differences and reject the null of non-stationarity.

(b) Now, consider the following equation, where the variables $\Delta\texttt{gold}_t$ and $\texttt{gold}_t$ are respectively the rate of change and the logarithm of an index of the market price of gold at time t, and figures in parentheses are standard errors.

$$\begin{aligned}\Delta\texttt{gold}_t &= \underset{(0.016)}{0.035} + \underset{(0.0003)}{0.0002}t - \underset{(0.018)}{0.017}\texttt{gold}_{t-1} \\ &\quad + \underset{(0.096)}{0.225}\Delta\texttt{gold}_{t-1} - \underset{(0.097)}{0.192}\Delta\texttt{gold}_{t-2} + \underset{(0.096)}{0.082}\Delta\texttt{gold}_{t-3}\end{aligned} \tag{8}$$

The equation was estimated using monthly data over the period from July 1976 to December 1985. Test (at the 5% level) the hypothesis that the logarithm of the gold price is a random walk with a drift. Clearly state the underlying alternative hypothesis.

Solution: This is the Augmented Dickey-Fuller test for $\texttt{gold}_t$. The number of observations is $T = 114$. The alternative hypothesis is that the deviations of log gold price from a linear trend are stationary. The 3 lags of the dependent variable used as explanatory variables on the right-hand side of the regression are meant to correct for potentially serial correlation in the error term, and hence makes sure that inference is not invalidated. The t-statistic of the coefficient on $\texttt{gold}_{t-1}$ is $t = -0.017/0.018 = -0.944$. From the Dickey-Fuller tables for τ_t (constant and trend), the 5% critical value for the statistic (assuming 100 observations, which is close enough) is -3.45. Hence we cannot reject the hypothesis that the market price of gold is a random walk with a drift.

S8.3 Engle and Granger at McDonald's.

We are interested in testing the theory of Absolute Purchasing Power Parity (PPP). If Absolute PPP holds then the real exchange rate equals one and so:

$$P_t = P_t^* e_t \tag{9}$$

where P_t and P_t^* are the price indices in the domestic and foreign economy respectively and e_t is the nominal exchange rate.

(a) Suppose e_t is non-stationary. Could Absolute PPP be true if both P_t and P_t^* were stationary? Carefully explain your answer.

Solution: Denote the real exchange rate by r_t and observe that it is easier to work with the logarithmic transformation of the underlying variables, $\ln r_t, \ln P_t, \ln P_t^*$ and

$\ln e_t$. Note that since the logarithm is a monotonic transformation, the stationarity property is preserved. Now, absolute PPP implies that the real exchange rate, $r_t = P_t^* e_t / P_t$ is equal to one. In the log-form this can be written as follows, where we recall that $\ln 1 = 0$:

$$\ln r_t = \ln P_t^* + \ln e_t - \ln P_t = 0.$$

or alternatively as

$$\ln e_t = \ln P_t - \ln P_t^* = \ln \left(\frac{P_t}{P_t^*} \right). \tag{10}$$

Equation (10) implies that if Absolute PPP holds, then the log of the nominal exchange rate must change as a one-to-one correspondence with the log of the relative price index. If $\ln e_t$ is non-stationary, then $\ln P_t - \ln P_t^*$ must also be non-stationary, in order for the left-hand side of the equation to balance with the right-hand side of the equation. Consequently, if both P_t and P_t^* are stationary, while e_t is non-stationary, then Absolute PPP does not hold.

(b) Describe how we can test whether P_t and P_t^* are non-stationary. Discuss the alternative specifications for the test setup and what considerations should determine which setup we adopt.

Solution: Stationarity of $p_t = \ln P_t$ can be tested using either the Dickey-Fuller (DF) or the Augmented Dickey-Fuller (ADF) test for the presence of a unit root. The difference between the two is that ADF contains lagged dependent variables on the right-hand side of the regression, in order to correct for potential serial correlation of the error terms, while DF assumes the errors are serially uncorrelated (think about Gauss-Markov assumptions). The test specification may also include deterministic components, such as an intercept and a trend, which should be included based on the visual assessment of the time series. Failure to specify the correct deterministic components (e.g. no intercept implies zero mean of the series), will invalidate inference about the unit root.

In order to conduct inference, start with the simplest unit-root DF setting without any deterministic components:

$$p_t = \phi p_{t-1} + \varepsilon_t$$

where ε_t is the error term (assumed serially uncorrelated). Test for a unit root using $H_0 \colon \phi = 1$, that is p_t contains a unit root (non-stationary), against $H_1 \colon \phi < 1$, that is the series is stationary. Equivalently, subtract p_{t-1} on both sides and obtain the DF specification in the first difference

$$\Delta p_t = \beta p_{t-1} + \varepsilon_t$$

where $\beta = \phi - 1$, so test $H_0 \colon \beta = 0$ against $H_1 \colon \beta < 0$, using the t-statistic for β, which has the Dickey-Fuller distribution and not $N(0,1)$ under the null. The critical values for the test, DF_α, at an arbitrary significance level α will differ depending on the deterministic components included in the specification. We reject the null of non-stationarity if $t_\beta < DF_\alpha$. The possible versions, including the ADF correction for potential serial correlation in ε_t, are given by

$$\Delta p_t = \beta p_{t-1} + \sum_{j=1}^{q} \theta_j \Delta p_{t-j} + \varepsilon_t$$

$$\Delta p_t = \alpha + \beta p_{t-1} + \sum_{j=1}^{q} \theta_j \Delta p_{t-j} + \varepsilon_t$$

$$\Delta p_t = \alpha + \delta t + \beta p_{t-1} + \sum_{j=1}^{q} \theta_j \Delta p_{t-j} + \varepsilon_t$$

where α is an intercept, δt is a time trend and q is the order (highest lag) of the dependent variable. As a rule of thumb, we should always include the intercept in our specification, in order to allow for non-zero mean of p_t. If the mean is indeed zero, then the estimate of α will be consistently estimated as zero anyway. If the trend is included, and the null of a unit root is not rejected, then p_t is a random walk with a drift (otherwise it is trend-stationary).

(c) Outline the main problems associated with hypotheses testing in time-series regressions that contain non-stationary variables.

Solution: When two (or more) non-stationary time series with different stochastic trends are regressed, this may produce seemingly significant regression results even if in reality (in population / the data-generating process) they are independent. This is called spurious regression: when conducting inference on the regression coefficient, we reject the null of insignificance more often than allowed by the nominal rejection (size) of the test given by the significance level α. As a simple example, consider regressing two unrelated random walks X_t and Y_t, as $Y_t = \beta X_t + u_t$, where u_t is the error term and clearly $\beta = 0$. However, in this case the usual LLN and CLT do not apply. In particular, since X_t is a random walk,

$$\hat{\beta} - \beta = \frac{\sum_{t=1}^{T} X_t u_t}{\sum_{t=1}^{T} X_t^2}$$

does not converge to zero, since $\sum X_t^2$ has infinite variance. Hence, using t_β to test for significance of β will yield the wrong result.

(d) If all the variables in (9) are $I(1)$, explain why the concept of cointegration is useful in determining the validity of Absolute PPP.

Solution: Two (or more) $I(1)$ series are called cointegrated when there exists an $I(0)$ linear relationship between them. This linear relationship can be interpreted as an equilibrium or a steady state. If Absolute PPP holds, then in the long-run the nominal interest rate should adjust in a one-to-one manner to the relative price change. However, we should also take into account the unobserved short-run effects, in particular we can think of the regression

$$\ln e_t = \beta \ln \left(\frac{P_t}{P_t^*} \right) + u_t$$

where u_t are those short-run shocks to the Absolute PPP and $\beta = 1$. If $\ln e_t$ is cointegrated with $\ln \left(\frac{P_t}{P_t^*} \right)$, u_t will be stationary around zero mean, and hence in the long-run the Absolute PPP will hold.

(e) Outline an equation you would estimate to test for cointegration. What restrictions would the theory of Absolute PPP imply on the cointegrating coefficients?

Solution: We can formulate our test of Absolute PPP in equation (10) as a cointegration test and use the Engle-Granger procedure. In particular, we should estimate the following restricted regression:

$$\ln e_t - \ln P_t + \ln P_t^* = u_t$$

where the coefficients are restricted to -1 and 1, and save the estimated residuals $\hat{u}_t$. Then, use the ADF to test for a unit root in $\hat{u}_t$. If we reject the null of non-stationarity in $\hat{u}_t$, conclude that the nominal exchange rate is cointegrated with the relative price change, and hence that Absolute PPP holds.

S8.4 Don't just partially adjust your consumption, error correct it!

An investigator wishes to examine the relationship between log consumption, c_t, and log disposable income, y_t using annual data for the period 1973-1996. He first estimates the so-called partial adjustment model, a dynamic relationship between the two variables, which takes the following form:

$$c_t = -0.70 + 0.86y_t + 0.21c_{t-1} + \varepsilon_t, \quad SSR = 0.005561. \tag{11}$$

(a) Suppose that the desired level of log consumption, c_t^*, is a linear function of the log disposable income:

$$c_t^* = \gamma_0 + \gamma_1 y_t. \tag{12}$$

In other words, the desired log consumption responds linearly and contemporaneously to log income. Show that regression (11) can be interpreted as the outcome of a process in which actual log consumption, c_t, adjusts partially to the desired level of log consumption, c_t^*. What is the adjustment speed implied by this equation? [Hint: use equations (11) and (12) to express Δc_t as a function of c_t^*.]

Solution: Subtract c_{t-1} from both sides of (11):

$$\begin{aligned} c_t - c_{t-1} &= -0.70 + 0.86y_t + (0.21 - 1)\, c_{t-1} + \varepsilon_t \\ \Delta c_t &= -0.70 + 0.86y_t - 0.79c_{t-1} + \varepsilon_t \\ \Delta c_t &= 0.79\,(-0.89 + 1.09y_t) - 0.79c_{t-1} + \varepsilon_t \end{aligned}$$

The desired consumption adjustment is given by $c_t^* = -0.89 + 1.09y_t$. Conclude

$$\Delta c_t = 0.79\,(c_t^* - c_{t-1}) + \varepsilon_t \tag{13}$$

$$c_t = 0.79c_t^* + 0.21c_{t-1} + \varepsilon_t \tag{14}$$

In this partial adjustment model, 79% of the adjustment is made in the first period.

(b) What is the long-run elasticity of consumption with respect to income?

Solution: From the equation above, the long-run elasticity is 1.09. Alternatively, note that the long-run equation in (log) levels, denoted $\bar{c}$ and $\bar{y}$, has the form

$$\bar{c} = -0.70 + 0.86\bar{y} + 0.21\bar{c} \implies \bar{c} = -0.89 + 1.09\bar{y}. \tag{15}$$

(c) Now, show that equation (11) can also be interpreted in terms of a model in which log consumption depends only on current and lagged values of log income. In terms of this model, what is the elasticity of current consumption with respect to an increase

in income two years earlier? [Hint: express equation (11) using the lag polynomial for c_t and hence express c_t as an infinite linear combination of the lags of y_t.]

Solution: Use the lag operator to express the original specification as

$$\begin{aligned} c_t &= -0.70 + 0.86y_t + 0.21Lc_t + \varepsilon_t \\ (1 - 0.21L)\, c_t &= -0.70 + 0.86y_t + \varepsilon_t \\ c_t &= (1 - 0.21L)^{-1} (-0.70 + 0.86y_t + \varepsilon_t) \end{aligned}$$

Note that the inverse of the lag polynomial converges:

$$\frac{1}{1 - 0.21L} = \sum_{i=0}^{\infty} 0.21^i L^i.$$

Therefore, we can write c_t as an infinite linear combination of the lags of y_t:

$$\begin{aligned} c_t &= -0.88 + 0.86 \sum_{i=0}^{\infty} 0.21^i L^i y_t + \sum_{i=0}^{\infty} 0.21^i L^i y_t c_t \\ &= -0.88 + 0.86 \sum_{i=0}^{\infty} 0.21^i y_{t-i} + \sum_{i=0}^{\infty} 0.21^i \varepsilon_{t-i} \end{aligned}$$

Note that the general expression can be written as

$$c_t = \gamma_0 + \gamma_1 \sum_{i=0}^{\infty} \varphi_i y_{t-i} + \sum_{i=0}^{\infty} \varphi_i \varepsilon_{t-i}. \tag{16}$$

The elasticity of current consumption with respect to an increase in income two years earlier is the coefficient on y_{t-2}, or $\varphi_2 = 0.86 \times 0.21^2 = 0.038$. The effect of income on consumption dies down to almost negligible after just two periods.

(d) The investigator now adds an additional explanatory variable, y_{t-1}, to the regression, with the following results

$$c_t = -0.28 + 0.97y_t + 0.66c_{t-1} - 0.61y_{t-1} + \varepsilon_t, \quad SSR = 0.003727 \tag{17}$$

Express this regression in the form of the Error Correction Model (ECM):

$$\Delta c_t = \theta_0 + \theta_1 \Delta y_t + \alpha\,(c_{t-1} - \beta y_{t-1}) + \varepsilon_t. \tag{18}$$

Interpret this specification with reference to short-run and long-run effects. Then, use this specification to compute the adjustment of c_t to a one period-shock to income (i.e. the coefficient on y_{t-1}). What is the long-run elasticity of consumption with respect to income?

Solution: Subtract c_{t-1} from both sides, and then subtract and add $0.97y_{t-1}$ from the right-hand side:

$$\begin{aligned} c_t &= -0.28 + 0.97y_t + 0.66c_{t-1} - 0.61y_{t-1} + \varepsilon_t \\ c_t - c_{t-1} &= -0.28 + 0.97\,(y_t - y_{t-1}) + (0.66 - 1)\, c_{t-1} + (0.97 - 0.61)\, y_{t-1} + \varepsilon_t \\ \Delta c_t &= -0.28 + 0.97\Delta y_t - 0.34c_{t-1} + 0.36y_{t-1} + \varepsilon_t \end{aligned}$$

$$\Delta c_t = -0.28 + 0.97\Delta y_t - 0.34\left(c_{t-1} - 1.06y_{t-1}\right) + \varepsilon_t \tag{19}$$

where $\theta_0 = -0.28$, $\theta_1 = 0.97$, $\alpha = -0.34$ and $\beta = 1.06$. The ECM splits the reaction of consumption to changes in income into the short-term and the long-term effects. The coefficient 0.97 on Δy_t is the contemporaneous short-run reaction of consumption to a change in income. The coefficient $\beta = 1.06$ is the long-run elasticity of consumption to income (note that this is slightly different from the long-run elasticity implied by the partial adjustment model). This suggests that there is a cointegrating relationship between consumption and income, where the co-movement is almost one-to-one. The deviations from the equilibrium, $\hat{u} = c_t - 1.06y_t$, is a stationary process. The speed of adjustment to the equilibrium is 0.34, suggesting that it takes on average 3 years for the deviation to adjust. The negative sign on this coefficient means that positive (negative) deviations are corrected downwards (upwards).

(e) Use a formal statistical test to assess whether the partial adjustment specification is an acceptable model of consumption for this data. [Hint: conduct an F-test to compare the ECM in (19) to the partial adjustment model in (11).]

Solution: We are comparing ECM in (19), the unrestricted model, to the partial adjustment model in (11), the restricted model:

$$\begin{aligned} c_t &= -0.28 + 0.97y_t + 0.66c_{t-1} - 0.61y_{t-1} + \varepsilon_t, \quad & SSR_u &= 0.003727 \\ c_t &= -0.70 + 0.86y_t + 0.21c_{t-1} + e_t, & SSR_r &= 0.005561 \end{aligned}$$

The null hypothesis is that the coefficient on y_{t-1} is zero. We can use an F-test with one restriction, $q = 1$, $T = 23$ observations, $k = 3$ explanatory variables in the unrestricted model, and so $T - (k+1) = 19$ degrees of freedom:

$$V = \frac{\left(SSR_r - SSR_u\right)/1}{SSR_u/19} \sim F_{1,19}.$$

Compute the observed test statistic, $v = 9.35$, and compare with the critical value at the 5% significance level, $F_{0.95,1,19} = 4.38$. We can reject the null hypothesis at 5%, and conclude that y_{t-1} is a relevant explanatory variable for c_t. There is evidence that the partial adjustment specification is not consistent with the data, and we should prefer the ECM.

S8.5 Prediction is very difficult, especially if it's about the future!
Suppose that the data are generated by the MA(1) model

$$y_t = \varepsilon_t - \frac{1}{2}\varepsilon_{t-1}, \tag{20}$$

where ε_t are iid standard normal random variables.

(a) Compute expectation, variance, autocovariance and autocorrelation functions for the process y_t.

Solution: This is an MA(1) process with the coefficient $\theta = -1/2$. Generally, an MA(1) process is expressed as

$$y_t = \varepsilon_t + \theta\varepsilon_{t-1}, \quad t = 1, \ldots, T, \tag{21}$$

where in our case $\varepsilon_t \sim \text{iid}N(0,1)$. Compute the first two moments of y_t using properties of the standard normal errors ε_t, namely $\mathbb{E}[\varepsilon_t] = 0$ and $\text{Var}(\varepsilon_t) = 1$:

$$\mathbb{E}[y_t] = \mathbb{E}[\varepsilon_t + \theta\varepsilon_{t-1}] = 0 \tag{22}$$

$$\text{Var}(y_t) = \text{Var}(\varepsilon_t + \theta\varepsilon_{t-1}) = \text{Var}(\varepsilon_t) + \theta^2\text{Var}(\varepsilon_{t-1}) = \left(1+\theta^2\right), \tag{23}$$

where for the variance we also used the independence of the error terms that implies $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for any $s \neq t$. We can also compute covariance of y_t with $y_{t-\tau}$ for displacement $|\tau| = 1$:

$$\begin{aligned}
\text{Cov}(y_t, y_{t-1}) &= \text{Cov}(\varepsilon_t + \theta\varepsilon_{t-1}, \varepsilon_{t-1} + \theta\varepsilon_{t-2}) \\
&= \text{Cov}(\varepsilon_t, \varepsilon_{t-1}) + \text{Cov}(\varepsilon_t, \theta\varepsilon_{t-2}) + \text{Cov}(\theta\varepsilon_{t-1}, \varepsilon_{t-1}) + \text{Cov}(\theta\varepsilon_{t-1}, \theta\varepsilon_{t-2}) \\
&= \theta\text{Cov}(\varepsilon_{t-1}, \varepsilon_{t-1}) \\
&= \theta\text{Var}(\varepsilon_{t-1}) \\
&= \theta\left(1+\theta^2\right)
\end{aligned}$$

where again we used $\text{Var}(\varepsilon_t) = 1$ and $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for any $s \neq t$. Notice that $\text{Cov}(y_t, y_{t-\tau}) = 0$ for any $|\tau| > 1$ and so the autocovariance and autocorrelation functions for a general MA(1) process are given by

$$\gamma(\tau) = \begin{cases} \left(1+\theta^2\right) & \text{for } \tau = 0 \\ \theta\left(1+\theta^2\right) & \text{for } |\tau| = 1 \\ 0 & \text{for } \tau > 1 \end{cases} \tag{24}$$

$$\rho(\tau) = \begin{cases} 1 & \text{for } \tau = 0 \\ \theta & \text{for } |\tau| = 1 \\ 0 & \text{for } \tau > 1 \end{cases} \tag{25}$$

For the special case in (20) with $\theta = -1/2$, we have

$$\mathbb{E}[y_t] = 0, \qquad \gamma(\tau) = \begin{cases} 5/4 & \tau = 0 \\ -5/8 & |\tau| = 1 \\ 0 & \tau > 1 \end{cases}, \qquad \rho(\tau) = \begin{cases} 1 & \tau = 0 \\ -1/2 & |\tau| = 1 \\ 0 & \tau > 1 \end{cases} \tag{26}$$

(b) Show that the MA(1) process in (20) is invertible and can be expressed as an AR(∞) process. [Hint: use the lag operator L to express ε_t as an infinite sum of lags of y_t.]

Solution: Use the lag operator L, which acts on a time series transforming it to its lag, for example $Ly_t = y_{t-1}$, or generally $L^n y_t = y_{t-n}$, to re-express the process as:

$$y_t = \varepsilon_t + \theta\varepsilon_{t-1} = \varepsilon_t + \theta L\varepsilon_t = (1+\theta L)\,\varepsilon_t \implies \varepsilon_t = (1+\theta L)^{-1} y_t. \tag{27}$$

Condition $|\theta| < 1$ implies that the infinite sum converges:

$$\frac{1}{1+\theta L} = \sum_{i=0}^{\infty} (-\theta)^i L^i.$$

Thus, ε_t can be represented as an AR(∞), an infinite sum of past realization of y_t:

$$\varepsilon_t = (1+\theta L)^{-1} y_t = \sum_{i=0}^{\infty} (-\theta)^i L^i y_t = \sum_{i=0}^{\infty} (-\theta)^i y_{t-i} = y_t - \theta y_{t-1} + \theta^2 y_{t-2} - \dots \tag{28}$$

Conclude that MA(1) is invertible whenever $|\theta| < 1$. In our special case with $\theta = -1/2$ this representation becomes

$$\varepsilon_t = \sum_{i=0}^{\infty} \left(\frac{1}{2}\right)^i y_{t-i}. \tag{29}$$

(c) Express ε_t as a function of $y_t, y_{t-1}, \ldots, y_1$ and ε_0.
Solution: Note that the process in (20) can be written as $\varepsilon_t = y_t - \theta\varepsilon_{t-1}$, and use backward substitution to derive the finite-time representation for ε_t:

$$\varepsilon_t = y_t - \theta\varepsilon_{t-1} = y_t - \theta\left(y_{t-1} - \theta\varepsilon_{t-2}\right) = \cdots = \theta^t\varepsilon_0 - \sum_{i=0}^{t-1} \theta^i y_{t-i}. \tag{30}$$

For $\theta = -1/2$ this becomes

$$\varepsilon_t = y_t + \frac{1}{2}y_{t-1} + \cdots + \left(\frac{1}{2}\right)^{t-1} y_1 + \left(\frac{1}{2}\right)^t \varepsilon_0. \tag{31}$$

Notice, that y_t in this setting has the following specificaiton

$$y_t = -\frac{1}{2}\left(y_{t-1} + \frac{1}{2}y_{t-2} + \cdots + \left(\frac{1}{2}\right)^{t-2} y_1\right) - \left(\frac{1}{2}\right)^t \varepsilon_0 + \varepsilon_t \tag{32}$$

(d) Suppose that you have data on $y_1, y_{2,} \ldots, y_T$ and you are told that $\varepsilon_0 = 0$. What are the optimal forecasts of y_{T+1} and y_{T+2}?
Solution: First, observe that using expression (32), y_T is given by

$$y_T = -\frac{1}{2}\left(y_{T-1} + \frac{1}{2}y_{T-2} + \cdots + \left(\frac{1}{2}\right)^{T-2} y_1\right) - \left(\frac{1}{2}\right)^T \varepsilon_0 + \varepsilon_T.$$

Now, denote the information set at time t by $\Omega_t = (\varepsilon_0, y_1, y_2, \ldots, y_t)$. Then the optimal forecast for y_{T+1} at time T, which we denote by $y_{T+1|T}$, is the conditional expectation of y_{T+1} given the information set Ω_T, in other words, $y_{T+1|T} = \mathbb{E}\left[y_{T+1}|\Omega_T\right]$. Use $\varepsilon_0 = 0$ and $\mathbb{E}\left[\varepsilon_t\right] = 0$ for all t to compute the following

$$\begin{aligned}
y_{T+1|T} &= \mathbb{E}\left[y_{T+1}|\Omega_T\right] \\
&= \mathbb{E}\left[-\frac{1}{2}\left(y_T + \frac{1}{2}y_{T-1} + \cdots + \left(\frac{1}{2}\right)^{T-1} y_1\right) - \left(\frac{1}{2}\right)\varepsilon_0 + \varepsilon_T \,\middle|\, \Omega_T\right] \\
&= \mathbb{E}\left[-\frac{1}{2}\left(y_T + \frac{1}{2}y_{T-1} + \cdots + \left(\frac{1}{2}\right)^{T-1} y_1\right) \,\middle|\, \Omega_T\right] \\
&= -\frac{1}{2}\left[y_T + \frac{1}{2}y_{T-1} + \cdots + \left(\frac{1}{2}\right)^{T-1} y_1\right].
\end{aligned}$$

The two-periods ahead optimal forecast, denoted $y_{T+2|T}$, will now depend not on y_{T+1},as it is unobserved at time T, but rather on the corresponding optimal forecast $y_{T+1|T}$, which we've just computed. Formally:

$$y_{T+2|T} = \mathbb{E}\left[y_{T+2}|\Omega_T\right]$$

$$= \mathbb{E}\left[-\frac{1}{2}\left(y_{T+1|T} + \frac{1}{2}y_T + \cdots + \left(\frac{1}{2}\right)^T y_1\right) - \left(\frac{1}{2}\right)\varepsilon_0 + \varepsilon_{T+1} \,\middle|\, \Omega_T\right]$$
$$= \mathbb{E}\left[-\frac{1}{2}\left(y_{T+1|T} + \frac{1}{2}y_T + \cdots + \left(\frac{1}{2}\right)^T y_1\right) \,\middle|\, \Omega_T\right]$$

Note that we can express the first term inside the expectation as

$$y_{T+1|T} = -\frac{1}{2}y_T - \left(\frac{1}{2}\right)^2 y_{T-1} - \cdots - \left(\frac{1}{2}\right)^T y_1$$

Therefore, the terms inside the expectation above cancel out:

$$y_{T+2|T} = -\frac{1}{2}\mathbb{E}\left[-\frac{1}{2}y_T - \cdots - \left(\frac{1}{2}\right)^T y_1 + \frac{1}{2}y_T + \cdots + \left(\frac{1}{2}\right)^T y_1 \,\middle|\, \Omega_T\right] = 0.$$

Hence, the optimal forecast for an MA(1) process y_t at any horizon longer than 1 period is zero, $y_{T+\tau|T} = 0$ for $\tau > 1$. This is equal to the unconditional mean of y_t, which we found in (a). Note that this is because the autocorrelation function for y_t is zero for any displacement $|\tau| > 1$.

(e) Consider an alternative forecast of y_{T+1}, which has form Cy_T, where C is some constant. What is the best choice of C?

Solution: To find C, consider minimizing the squared forecast error:

$$\mathbb{E}\left[(y_{T+1} - Cy_T)^2\right] = \mathbb{E}\left[y_{T+1}^2\right] - 2C\mathbb{E}\left[y_{T+1}y_T\right] + C^2\mathbb{E}\left[y_T^2\right]$$

Recall that $\mathbb{E}\left[\varepsilon_t^2\right] = 1$ and $\mathbb{E}\left[\varepsilon_t\varepsilon_s\right] = 0$, and use this to find the first term above:

$$\mathbb{E}\left[y_{T+1}^2\right] = \mathbb{E}\left[\left(\varepsilon_{T+1} - \frac{1}{2}\varepsilon_T\right)^2\right] = \mathbb{E}\left[\varepsilon_{T+1}^2\right] - \mathbb{E}\left[\varepsilon_{T+1}\varepsilon_T\right] + \frac{1}{4}\mathbb{E}\left[\varepsilon_T^2\right] = \frac{5}{4}.$$

Consider the second term

$$\mathbb{E}\left[y_{T+1}y_T\right] = \mathbb{E}\left[\left(\varepsilon_{T+1} - \frac{1}{2}\varepsilon_T\right)\left(\varepsilon_T - \frac{1}{2}\varepsilon_{T-1}\right)\right]$$
$$= \mathbb{E}\left[\varepsilon_{T+1}\varepsilon_T\right] - \frac{1}{2}\mathbb{E}\left[\varepsilon_{T+1}\varepsilon_{T-1}\right] - \frac{1}{2}\mathbb{E}\left[\varepsilon_T\varepsilon_T\right] + \frac{1}{4}\mathbb{E}\left[\varepsilon_T\varepsilon_{T-1}\right] = -\frac{1}{2}$$

Consider the third term

$$\mathbb{E}\left[y_T^2\right] = \mathbb{E}\left[\left(\varepsilon_T - \frac{1}{2}\varepsilon_{T-1}\right)^2\right] = \mathbb{E}\left[\varepsilon_T^2\right] - \mathbb{E}\left[\varepsilon_T\varepsilon_{T-1}\right] + \frac{1}{4}\mathbb{E}\left[\varepsilon_{T-1}^2\right] = \frac{5}{4}.$$

Putting these results together we get

$$\mathbb{E}\left[(y_{T+1} - Cy_T)^2\right] = \frac{5}{4} - C + \frac{5}{4}C^2.$$

Minimize this with respect to C by setting the first-order derivative to zero to conclude that $C = 2/5$. Recall that the MA(1) coefficient was $\theta = -1/2$ and the variance of y_t was $\mathrm{Var}\left(y_t\right) = 5/4$, and so observe that $C = -\theta/\mathrm{Var}\left(y_t\right)$.

S8.6 If AR(1) and a measurement error had a baby, they'd call it ARMA(1,1). Consider an AR(1) process

$$x_t = 0.5x_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{iid}\, N(0,1). \tag{33}$$

Suppose that we observe a noisy version of x_t, namely

$$y_t = x_t + \eta_t, \tag{34}$$

where η_t is a white noise process with variance 1, independent of the x_t series.

(a) Find the population autocovariance function for x_t.

Solution: First, note that x_t is an AR(1) process $x_t = \phi x_{t-1} + \varepsilon_t$, with $\phi = 0.5$ and $\text{Var}(\varepsilon_t) = 1$. It is obvious that $\text{Cov}(x_{t-\tau}, \varepsilon_t) = 0$ for $\tau > 0$. Then, since the process x_t is stationary, its variance, $\gamma_x(0)$, is constant in time, and its autocovariance $\gamma_x(\tau)$ depends only on the displacement parameter τ, not on time, compute

$$\gamma_x(0) = \text{Var}(x_t) = \text{Var}(\phi x_t + \varepsilon_t) = \phi^2 \text{Var}(x_t) + \text{Var}(\varepsilon_t) \implies \gamma_x(0) = \left(\frac{1}{1-\phi^2}\right)$$

$$\gamma_x(1) = \text{Cov}(x_t, x_{t-1}) = \text{Cov}(\phi x_{t-1} + \varepsilon_t, x_{t-1}) = \phi \text{Var}(x_{t-1}) = \phi\left(\frac{1}{1-\phi^2}\right)$$

$$\gamma_x(2) = \text{Cov}(x_t, x_{t-2}) = \text{Cov}\left(\phi^2 x_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t, x_{t-2}\right) = \phi^2 \text{Var}(x_{t-2}) = \phi^2\left(\frac{1}{1-\phi^2}\right)$$

Using the recursive property of the autocovariance function, deduce that for $\tau \geq 0$

$$\gamma_x(\tau) = \phi^\tau \left(\frac{1}{1-\phi^2}\right). \tag{35}$$

Since in our case $\phi = 0.5$ we can conclude that for $\tau \geq 0$

$$\gamma_x(\tau) = \left(\frac{1}{2}\right)^\tau \left(\frac{4}{3}\right). \tag{36}$$

(b) Find the population autocovariance function for y_t.

Solution: Since η_t are independent of x_t, $\text{Cov}(\eta_s, x_t) = 0$, for all s, t, we compute

$$\begin{aligned}\text{Cov}(y_t, y_{t-\tau}) &= \text{Cov}(x_t + \eta_t, x_{t-\tau} + \eta_{t-\tau}) \\ &= \text{Cov}(x_t, x_{t-\tau}) + \text{Cov}(x_t, \eta_{t-\tau}) + \text{Cov}(\eta_t, x_{t-\tau}) + \text{Cov}(\eta_t, \eta_{t-\tau}) \\ &= \text{Cov}(x_t, x_{t-\tau}) + \text{Cov}(\eta_t, \eta_{t-\tau}).\end{aligned}$$

Now, compute the autocovariance function for the process y_t, starting with the variance, and recall that we assumed $\text{Cov}(\eta_t, \eta_t) = 1$:

$$\gamma_y(0) = \text{Var}(y_t) = \text{Cov}(x_t, x_t) + \text{Cov}(\eta_t, \eta_t) = \left(\frac{1}{1-\phi^2}\right) + 1 = \left(\frac{2-\phi^2}{1-\phi^2}\right). \tag{37}$$

Since the error terms are independent, so that $\text{Cov}(\eta_t, \eta_{t-\tau}) = 0$, for $\tau > 0$, the general expression for the autocovariance function of y_t for $\tau > 0$ is

$$\gamma_y(\tau) = \text{Cov}(y_t, y_{t-\tau}) = \text{Cov}(x_t, x_{t-\tau}) = \phi^\tau \left(\frac{1}{1-\phi^2}\right). \tag{38}$$

To summarize the general expression for the autocovariance function of y_t is given

$$\gamma_y(\tau) = \begin{cases} \left(\dfrac{2-\phi^2}{1-\phi^2}\right) & \text{for } \tau = 0 \\ \\ \phi^\tau \left(\dfrac{1}{1-\phi^2}\right) & \text{for } \tau > 0 \end{cases} \tag{39}$$

For the special case in this questions with $\phi = 0.5$, the autocovariance function is

$$\gamma_y(\tau) = \begin{cases} \dfrac{7}{3} & \text{for } \tau = 0 \\ \\ \left(\dfrac{1}{2}\right)^\tau \left(\dfrac{4}{3}\right) & \text{for } \tau > 0 \end{cases} \tag{40}$$

(c) Now, consider a stationary, stable and invertible, ARMA(1,1) process

$$z_t = \varphi z_{t-1} + u_t + \theta u_{t-1}, \quad u_t \sim \text{iid}\, N\left(0, \sigma^2\right). \tag{41}$$

Show that the variance and the first-order autocovariance of z_t are given by

$$\text{Var}(z_t) = \left[\frac{1+\theta^2+2\varphi\theta}{1-\varphi^2}\right]\sigma^2 \tag{42}$$

$$\text{Cov}(z_t, z_{t-1}) = \left[\frac{(\varphi+\theta)(1+\varphi\theta)}{(1-\varphi^2)}\right]\sigma^2. \tag{43}$$

Solution: First, note that since z_t is stationary, its variance is time-invariant $\text{Var}(z_t) = \text{Var}(z_{t-1})$. Then, use (41) to compute

$$\begin{aligned} \text{Var}(z_t) &= \text{Var}(\varphi z_{t-1} + u_t + \theta u_{t-1}) \\ &= \varphi^2 \text{Var}(z_{t-1}) + \text{Var}(u_t) + \theta^2 \text{Var}(u_{t-1}) \\ &\quad +2\Big[\varphi \text{Cov}(z_{t-1}, u_t) + \varphi \text{Cov}(z_{t-1}, u_{t-1}) + \theta \text{Cov}(u_t, u_{t-1})\Big] \end{aligned}$$

Recall that $\text{Var}(u_t) = \text{Var}(u_{t-1}) = \sigma^2$ and $\text{Cov}(u_t, u_{t-1}) = 0$. Also, since z_{t-1} does not depend on u_t, as suggested by equation (41), $\text{Cov}(z_{t-1}, u_t) = 0$. Finally,

$$\text{Cov}(z_{t-1}, u_{t-1}) = \text{Cov}(\varphi z_{t-2} + u_{t-1} + \theta u_{t-2}, u_{t-1}) = \text{Cov}(u_{t-1}, u_{t-1}) = \sigma^2.$$

Substitute these pieces into the expression for $\text{Var}(z_t)$ and re-arrange:

$$\begin{aligned} &\text{Var}(z_t) - \varphi^2 \text{Var}(z_t) = \left(1+\theta^2\right)\sigma^2 + 2\varphi\sigma^2 \\ &\implies \gamma_z(0) = \text{Var}(z_t) = \left[\frac{1+\theta^2+2\varphi\theta}{1-\varphi^2}\right]\sigma^2. \end{aligned}$$

Note the autocovariance function notation $\gamma(\tau)$ at displacement τ. Similarly, compute $\gamma_z(1) = \text{Cov}(z_t, z_{t-1})$:

$$\text{Cov}(z_t, z_{t-1}) = \text{Cov}(\varphi z_{t-1} + u_t + \theta u_{t-1}, z_{t-1})$$

$$
\begin{aligned}
&= \varphi \mathrm{Cov}\left(z_{t-1}, z_{t-1}\right) + \mathrm{Cov}\left(u_t, z_{t-1}\right) + \theta \mathrm{Cov}\left(u_{t-1}, z_{t-1}\right) \\
&= \left[\frac{\varphi\left(1+\theta^2+2\varphi\theta\right)}{1-\varphi^2}\right]\sigma^2 + \theta\sigma^2 \\
&= \left[\frac{\varphi+\varphi\theta^2+2\varphi^2\theta+\theta-\varphi^2\theta}{1-\varphi^2}\right]\sigma^2 \\
&= \left[\frac{(\varphi+\theta)(1+\varphi\theta)}{1-\varphi^2}\right]\sigma^2
\end{aligned}
$$

(d) Use the above expressions for $\mathrm{Var}\left(z_t\right)$ and $\mathrm{Cov}\left(z_t, z_{t-1}\right)$ to show that the population autocovariance function for z_t is given by

$$
\gamma_z(\tau) = \begin{cases} \left[\dfrac{1+\theta^2+2\varphi\theta}{1-\varphi^2}\right]\sigma^2 & \text{for } \tau = 0 \\ \\ \varphi^{\tau-1}\left[\dfrac{(\varphi+\theta)(1+\varphi\theta)}{1-\varphi^2}\right]\sigma^2 & \text{for } \tau > 0 \end{cases} \tag{44}
$$

Solution: Take covariance of z_t with $z_{t-\tau}$ on both sides of (41):

$$
\begin{aligned}
\mathrm{Cov}\left(z_t, z_{t-\tau}\right) &= \mathrm{Cov}\left(\varphi z_{t-1} + u_t + \theta u_{t-1}, z_{t-\tau}\right) \\
&= \varphi \mathrm{Cov}\left(z_{t-1}, z_{t-\tau}\right) + \mathrm{Cov}\left(u_t, z_{t-\tau}\right) + \theta \mathrm{Cov}\left(u_{t-1}, z_{t-\tau}\right)
\end{aligned}
$$

Recall that as z_t is stable, it admits an MA(∞) representation, that is z_t can be expressed as a linear combination of $u_t, u_{t-1}, u_{t-2}, \ldots$, while $z_{t-\tau}$ is a linear combination of $u_{t-\tau}, u_{t-\tau-1}, u_{t-\tau-2}, \ldots$. Therefore, $\mathrm{Cov}\left(u_t, z_{t-\tau}\right) = 0$ and $\mathrm{Cov}\left(u_{t-1}, z_{t-\tau}\right) = 0$, for any $\tau > 0$. Conclude that for $\tau > 0$

$$
\mathrm{Cov}\left(z_t, z_{t-\tau}\right) = \varphi \mathrm{Cov}\left(z_{t-1}, z_{t-\tau}\right). \tag{45}
$$

Since this equation is recursive, and the autocovariance function of a stationary process z_t only depends on the displacement parameter τ, and not on the time period t, we can write the Yule-Walker equations for the autocovariance function $\gamma_z(\tau)$ as

$$
\gamma_z(\tau) = \varphi\gamma_z(\tau-1) = \cdots = \varphi^{\tau-1}\gamma_z(1), \quad \text{for } \tau > 0 \tag{46}
$$

As we know the values of $\gamma_z(0)$ and $\gamma_z(1)$ from expressions (42) and (43), and as the Yule-Walker equation is recursive, the autocovariance function of z_t follows and takes the form in (44).

S8.7 Granger (alone) at KFC.

Consider the dynamic model for $t = 1, 2, \ldots$

$$
y_t = y_{t-1} + \beta x_{t-1} + \varepsilon_t, \tag{47}
$$

$$
x_t = \gamma y_t + v_t, \tag{48}
$$

where y_0, x_0 are random variables drawn from a distribution with zero mean, finite variances and covariance σ_{xy}, and ε_t, v_t are independent white noise processes with variance σ_ε^2 and σ_v^2, respectively. Finally, assume that

$$
\mathbb{E}\left[\varepsilon_t \mid y_{t-1}, x_{t-1}, y_{t-2}, x_{t-2}, \ldots\right] = \mathbb{E}\left[v_t \mid y_{t-1}, x_{t-1}, y_{t-2}, x_{t-2}, \ldots\right] = 0
$$

(a) What is the necessary condition for the process y_t to be stationary?
Solution: Substitute (48) into (47) to obtain

$$y_t = (1 + \beta\gamma)\, y_{t-1} + (\beta v_{t-1} + \varepsilon_t)\,,$$

Observe that $(1 + \beta\gamma)\, y_{t-1}$ is the AR part of y_t and $(\beta v_{t-1} + \varepsilon_t)$ is the MA part. The MA part is stationary by definition, since v_t and ε_t are two independent white noise processes. The AR part will be stable (and hence y_t stationary) if the root of the AR lag polynomial lies outside the unit circle. The root of the lag polynomial is given by

$$\Phi(z) = (1 - (1 + \beta\gamma)\, z) = 0 \implies z = (1 + \beta\gamma)^{-1}$$

The condition for stationarity is then $|z| > 1$, which is the case when $|(1 + \beta\gamma)| < 1$. This is satisfied when $-2 < \beta\gamma < 0$.

(b) What does it mean for a time-series process to be $I(1)$?
Solution: A time-series process z_t is integrated of order 1, or $I(1)$, if the AR lag polynomial $\Phi(L)$ of z_t, where $\Phi(L)\, z_t = \theta(L)\, u_t$, has a unit root, that is one root of $\Phi(z) = 0$ is $z = 1$. In this case, z_t can be expressed as a sum of white noise processes $u_t \sim \text{iit}\left(0, \sigma_u^2\right)$ or a random walk:

$$z_t = z_0 + \sum_{i=1}^{t} u_i.$$

If $z_t \sim I(1)$, then $\Delta z_t \sim I(0)$, so z_t is difference stationary.

(c) What is the order of integration of x_t if $\gamma = 0$?
Solution: If $\gamma = 0$, $x_t = v_t \sim I(0)$ is a white noise process.

(d) What is the order of integration of y_t and x_t if $\beta = 0$ and $\gamma \neq 0$? Are y_t and x_t cointegrated? If so, what is the cointegration coefficient?
Solution: If $\beta = 0$, $y_t = y_{t-1} + \varepsilon_t$ and $\Phi_y(L) = 1 - L$ has a unit root. Hence, using backward substitution, $y_t = y_0 + \sum_{i=1}^{t} \varepsilon_i \sim I(1)$ and is a random walk. We also call $\sum_{i=1}^{t} \varepsilon_i$ the stochastic trend of y_t. Since $x_t = \gamma y_t + v_t$, and $\gamma \neq 0$, then x_t has the same order of integration as y_t, and is also $I(1)$. Observe that in this case

$$x_t = y_0 + \gamma \sum_{i=1}^{t} \varepsilon_i + v_t$$

The stochastic trend of x_t accumulates the same random shocks as y_t, $\sum_{i=1}^{t} \varepsilon_i$, but is multiplied by γ. Clearly, y_t and x_t are cointegrated with coefficient γ.

(e) Explain the concept of Granger causality. Give conditions in terms of the values of β and γ under which each of the processes y_t, x_t Granger cause the other one.
Solution: x_t is said to Granger cause y_t if

$$\mathbb{E}\left[y_t | y_{t-1}, y_{t-2}, \ldots, x_{t-1}, x_{t-2} \ldots\right] \neq \mathbb{E}\left[y_t | y_{t-1}, y_{t-2}, \ldots\right]$$

For our particular case, consider

$$\mathbb{E}\left[y_t \mid y_{t-1}, x_{t-1}\right] = \mathbb{E}\left[y_{t-1} + \beta x_{t-1} + \varepsilon_t \mid y_{t-1}, x_{t-1}\right] = y_{t-1} + \beta x_{t-1}.$$

This means x_t Granger causes y_t if and only if $\beta \neq 0$. Similarly we can show that

$$\mathbb{E}\left[x_t \mid y_{t-1}, x_{t-1}\right] = \mathbb{E}\left[\gamma y_{t-1} + \gamma\beta x_{t-1} + \gamma\varepsilon_t + v_t \mid y_{t-1}, x_{t-1}\right] = \gamma y_{t-1} + \gamma\beta x_{t-1}$$

and y_t Granger causes x_t if and only if $\gamma \neq 0$.

(f) Assume that y_t and x_t are stationary. Prove that σ_{xy} is proportional to γ.
Solution: First, observe that $\mathbb{E}\left[y_t\right] = \mathbb{E}\left[x_t\right] = 0$, and so

$$\sigma_{xy} = \mathrm{Cov}\left(x_t, y_t\right) = \mathbb{E}\left[x_t y_t\right] - \mathbb{E}\left[x_t\right]\mathbb{E}\left[y_t\right] = \mathbb{E}\left[x_t y_t\right].$$

Use the fact that ε_t and v_t are zero mean processes independent of each other, so that $\mathbb{E}\left[\varepsilon_t v_t\right] = 0$, to compute

$$\sigma_{xy} = \mathbb{E}\left[x_t y_t\right] = \mathbb{E}\left[\left(\gamma y_t + v_t\right) y_t\right] = \gamma\mathbb{E}\left[y_t^2\right] + \mathbb{E}\left[v_t y_t\right] = \gamma\sigma_y^2$$

In this derivation we used the facts that

$$\mathbb{E}\left[v_t y_t\right] = \mathbb{E}\left[v_t\left(y_{t-1} + \beta x_{t-1} + \varepsilon_t\right)\right] = \mathbb{E}\left[v_t y_{t-1}\right] + \beta\mathbb{E}\left[v_t x_{t-1}\right] + \mathbb{E}\left[v_t\varepsilon_t\right] = 0$$

and that

$$\mathrm{Var}\left(y_t\right) = \mathbb{E}\left[y_t^2\right] - \left(\mathbb{E}\left[y_t\right]\right)^2 = \mathbb{E}\left[y_t^2\right] = \sigma_y^2$$

S8.8 Norman Borlaug singing in the rain.
`STATA` file `wheatIndia.dta` contains annual data on India's wheat production and rainfall from 1960 to 2016. Variable `wheat` measures wheat production in 1000 tonnes, variable `rain` measures average rainfall in mm, in the previous year across three major wheat-producing regions of India. Wheat in India is harvested from February to May, so the precipitation in the previous year is an important determinant of the harvest.

(a) Estimate the following model using OLS.

$$\log\left(\texttt{wheat}_t\right) = \alpha + \beta\log\left(\texttt{rain}_t\right) + u_t. \tag{49}$$

Discuss your results linking your discussion to the Gauss-Markov assumptions.
Solution: The OLS estimated regression, with standard errors in parenthesis:

$$\log\widehat{\left(\texttt{wheat}_t\right)} = \underset{(2.75)}{18.22} - \underset{(0.41)}{1.15}\log\left(\texttt{rain}_t\right), \quad R^2 = 0.12. \tag{50}$$

The negative sign of the slope coefficient is counter-intuitive. One would expect that more precipitation would yield better harvest of wheat. In this regression, the strict exogeneity assumption is not satisfied because the error term, u_t, contains trending omitted variables (such as technology, the area of wheat fields, etc). The expected value of these variables increases over time, as can be seen in Figure 1. Hence, the regression estimates may be biased and inconsistent. In fact, in this example the $\log\left(\texttt{rain}_t\right)$ variable is somewhat negatively trending. The negative coefficient on $\log\left(\texttt{rain}_t\right)$ is therefore an artifact of the opposite direction of the trends in $\log\left(\texttt{wheat}_t\right)$ and $\log\left(\texttt{rain}_t\right)$. This confounds the causal effect of rain on wheat.

(b) Make a time series plot of $\Delta \log(\texttt{wheat}_t)$, where Δ is the first-differencing operator. How would you characterize the time series $\Delta \log(\texttt{wheat}_t)$?

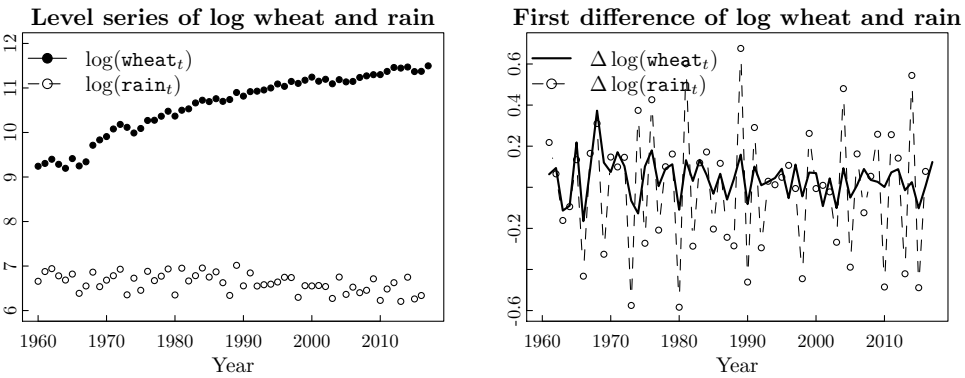

Figure 1: Level series (left panel) and first differences (right panel).

Solution: Looking at Figure 1, there are no obvious violations of stationarity, except, perhaps, some decrease in variance over time.

(c) Now consider the following model

$$\Delta \log(\texttt{wheat}_t) = \gamma + \delta \Delta \log(\texttt{rain}_t) + \varepsilon_t. \tag{51}$$

Estimate it using OLS. Discuss your results and compare them to those in (a). Which results you would trust more and why?

Solution: The OLS estimated regression, with standard errors in parenthesis:

$$\Delta \log \widehat{(\texttt{wheat}_t)} = \underset{(0.01)}{0.04} + \underset{(0.036)}{0.18}\, \Delta \log(\texttt{rain}_t), \quad R^2 = 0.32. \tag{52}$$

Now the estimate of the coefficient is positive, which accords with intuition. The intercept is saying that in the absence of changes in precipitation, the growth rate of wheat production in India was on average 4% per year. The coefficient 0.18 estimates the elasticity of wheat production with respect to rain: a 1% increase in precipitation causes 0.18% increase in wheat production. I would trust more these results than those in (a). In (c) the trend in wheat production has been eliminated by the first differencing.

(d) Test for no first-order serial correlation in ε_t using the Durbin-Watson test. Discuss the validity of this test. Run a residual-based test for no AR(1) serial correlation in ε_t and report the results.

Solution: Use the following `STATA` command for the Durbin-Watson test

```
>> estat dwatson
```

The DW-test statistic is equal to 2.088. The critical value for the DW-test has 2 explanatory variables in the original regression (a constant and a slope), and 56 observations. The value of the test statistic is larger than both the lower critical

value 1.528 and the upper critical value 1.601. Hence, we cannot reject the null of no serial correlation in favour of positive serial correlation. The test is valid under the assumptions of strict exogeneity, $\mathbb{E}\left[\varepsilon_t \middle| \log(\texttt{rain}_1), \ldots, \log(\texttt{rain}_T)\right] = 0$ for all $t = 1, \ldots, T$, and normality of errors. Here, the strict exogeneity is, perhaps, satisfied (no reason to believe that the errors at any time are related to the change in rain). However, normality of the errors might not hold, at least, no clear reason why it should. The residual-based test is the t-test in the regression of the fitted residuals $\hat{\varepsilon}_t$ from equation (52) on its own lag (under strict exogeneity). We have

$$\hat{\varepsilon}_t = 0.0003 - \underset{(0.137)}{0.048}\hat{\varepsilon}_{t-1}.$$

The t-statistic is -0.35, which is smaller than 1.96 by absolute value, and we cannot reject the null of no serial correlation at 5% significance.

(e) A friend says that since `rain` measures the amount of rain in the previous year, there is a possibility of "feedback" in the regression from (c). How would you interpret such a feedback, and what may its consequences be?

Solution: It is possible that the rain during January still has an effect on the February-May harvest of wheat in the same year. Therefore, it is possible that $\texttt{rain}_{t+1}$ (rain during the current year) is correlated to $\texttt{wheat}_t$. A consequence of such a feedback mechanism would be the inconsistency of the causal effect of rain on wheat harvest in the first-differenced regression.

(f) Propose a way to test the absence of the "feedback" that your friend is worried about. Perform the test that you propose.

Solution: One way to test for the absence of the feedback is to run an OLS regression of $\Delta \log(\texttt{rain}_t)$ on its own lag and on $\Delta \log(\texttt{whear}_{t-1})$. Then, test that the coefficient on $\Delta \log(\texttt{wheat}_{t-1})$ is zero using the t-test. We have

$$\Delta\widehat{\log(\texttt{rain}_t)} = -0.014 - 0.68\Delta \log(\texttt{rain}_{t-1}) - \underset{(0.38)}{0.02}\,\Delta \log(\texttt{wheat}_{t-1}).$$

The t-statistic is -0.05. We cannot reject the null of no feedback at any conceivable significance level. Note that this test is a special case of the Granger causality test, where we did not reject the hypothesis that changes in wheat production does not Granger-cause changes in rain.